

LINCOLN, CA, USA

WMO: 720614

Lat: 38.909N

Lon: 121.351W

Elev: 37

StdP: 100.88

Time Zone: -8.00 (NAP)

Period: 09-19

WBAN: 00205

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-2.3	-1.0	-7.2	2.1	9.5	-5.1	2.5	10.6	11.8	11.3	10.6	12.0	1.3	80	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	38.7	19.3	37.2	19.0	35.3	18.5	20.9	35.1	20.1	34.3	19.4	33.3	2.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	11.6	23.0	15.1	10.8	23.4	13.9	10.0	22.2	60.2	35.1	57.5	33.9	54.9	33.0	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	7.9	6.8	-4.9	41.0	1.8	1.1	-6.2	41.8	-7.3	42.5	-8.3	43.1	-9.6	43.9		
(5)			-5.2	22.1	1.8	1.0	-6.5	22.8	-7.5	23.4	-8.6	23.9	-9.9	24.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	8.0	9.9	12.2	14.7	18.1	23.1	25.8	24.8	22.2	16.9	11.0	7.6		
	DBStd	6.95	2.52	2.66	2.63	3.17	3.11	3.56	2.42	2.26	3.03	2.94	3.07	2.93		
	HDD10.0	227	71	33	10	2	0	0	0	0	0	0	24	86		
	HDD18.3	1529	319	237	191	114	42	5	0	0	5	63	221	332		
	CDD10.0	2498	10	29	77	144	252	392	488	457	367	215	53	12		
	CDD18.3	759	0	0	0	6	35	147	230	199	123	19	0	0		
	CDH23.3	11129	0	0	21	203	706	2188	3198	2635	1721	443	14	0		
	CDH26.7	6022	0	0	1	48	276	1234	1919	1499	906	138	0	0		
(14) Wind	WSAvg	2.7	2.2	3.0	3.0	2.9	3.2	3.0	3.1	2.8	2.5	2.3	2.1	2.3		
(15) Precipitation	PrecAvg	623	120	106	89	48	27	8	1	1	3	32	61	126		
	PrecMax	1019	325	304	286	170	115	40	3	14	13	108	134	289		
	PrecMin	249	6	0	1	1	0	0	0	0	0	2	5	14		
	PrecStd	177	87	74	81	37	29	12	1	3	4	28	37	79		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.9	22.2	26.1	31.0	34.9	40.9	40.9	40.0	37.8	33.0	25.2	17.8		
		MCWB	10.2	13.8	15.3	16.6	17.3	20.5	19.7	19.4	18.3	16.4	14.8	12.2		
	2%	DB	17.3	20.0	22.9	27.7	32.3	37.9	38.0	37.3	36.0	30.2	22.7	16.2		
		MCWB	10.3	12.6	13.5	16.0	16.7	19.6	19.2	18.9	18.2	15.9	13.4	11.0		
	5%	DB	15.2	17.7	22.0	26.0	30.1	36.0	37.2	35.9	33.9	27.6	20.9	14.9		
		MCWB	10.5	11.7	13.8	15.5	16.0	18.7	19.1	18.7	17.6	14.7	12.7	10.7		
	10%	DB	13.0	16.2	19.2	22.7	27.4	33.0	35.9	33.9	32.3	26.0	17.7	12.9		
		MCWB	9.8	11.0	12.5	14.1	15.2	17.6	18.8	18.4	17.3	14.4	11.4	9.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.6	14.9	16.5	19.1	19.0	22.0	22.2	21.1	20.2	18.8	16.3	14.3		
		MCDB	16.1	18.3	24.1	28.0	31.5	35.9	36.8	35.3	34.1	26.0	23.5	16.5		
	2%	WB	12.2	14.0	15.2	17.2	17.8	20.8	21.1	20.2	19.2	17.3	15.0	13.1		
		MCDB	14.5	18.0	20.9	25.0	29.4	34.9	35.6	34.6	33.2	26.4	20.6	15.3		
	5%	WB	11.3	12.9	14.1	15.9	16.7	19.9	20.2	19.5	18.6	16.4	13.7	12.1		
		MCDB	13.3	16.7	19.3	23.9	27.6	33.7	34.7	33.8	31.9	24.9	18.0	13.3		
	10%	WB	10.5	12.0	13.2	15.0	15.8	18.8	19.2	18.8	17.8	15.4	12.6	11.1		
		MCDB	12.7	15.2	18.2	22.3	26.1	31.3	33.6	32.4	30.2	23.5	16.2	12.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	11.7	12.5	14.4	15.9	18.2	18.9	18.6	18.2	16.6	13.5	11.0		
		MCDBR	14.4	15.1	17.7	19.2	21.0	21.7	20.9	20.5	21.5	20.2	18.2	12.7		
	5% WB	MCWBR	8.9	8.8	9.2	9.2	7.8	6.9	7.2	7.1	7.6	8.4	9.4	7.7		
		MCWBR	10.3	12.2	14.4	17.2	18.9	20.4	19.8	19.1	19.4	16.4	14.1	7.7		
(40) Clear-Sky Solar Irradiance	taub		0.323	0.326	0.351	0.358	0.374	0.366	0.364	0.347	0.357	0.349	0.328	0.338		
	taud		2.506	2.505	2.419	2.399	2.376	2.407	2.405	2.442	2.425	2.443	2.492	2.441		
	Ebn at Noon		853	901	907	919	905	909	908	918	889	859	837	799		
	Edh at Noon		78	88	106	115	120	117	116	109	104	93	78	78		
(44) All-Sky Solar Radiation	RadAvg		2.17	3.15	4.49	5.98	7.32	8.25	8.24	7.37	5.95	4.17	2.64	1.91		
	RadStd		0.39	0.42	0.51	0.55	0.48	0.35	0.26	0.24	0.22	0.29	0.21	0.33		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	+0.41	N/A	-1.11	N/A	N/A	N/A	N/A	-111	+139	+45

Nomenclature: See separate page